

Name:

Class:

Date:

Lab 13: Dice Game: Random & Libraries

Python Lab · Section 3.14–3.15 Libraries & Random Values



Goal

Import a library and use random values to build a dice game.

Warm-up: Predict

```
import random
print(random.randint(1, 6))
print(random.choice(["red", "blue"]))
```

What values can each line print? Run it 3 times.

AP Connection

- `random.randint(a, b) ≈ RANDOM(a, b)` — both include a and b.
- `import` uses a library; its documentation is the API (3.14).

Build It

Step 1. Roll two dice and print the sum.

```
import random
d1 = random.randint(1, 6)
d2 = random.randint(1, 6)
print(d1, "+", d2, "=", d1 + d2)
```

Step 2. Game rule: roll until you get doubles. Count the rolls.

```
rolls = 0
while True:
    d1 = random.randint(1, 6)
    d2 = random.randint(1, 6)
    rolls += 1
    if d1 == d2:
        break
print("Doubles after", rolls, "rolls")
```

Step 3. Weighted event: a 1-in-10 chance of a "golden die".

```
if random.randint(1, 10) == 1:
    print("GOLDEN DIE!")
```

Step 4. Read the documentation for `random.shuffle` and use it to shuffle a playlist.

```
songs = ["A", "B", "C", "D"]
random.shuffle(songs)
print(songs)
```

Date:

Build It (continued) · Level Up

What information did you need from the documentation to use random.shuffle?